BI:TASK-03:Exploratory Data Analysis (EDA) using Bank Marketing Dataset

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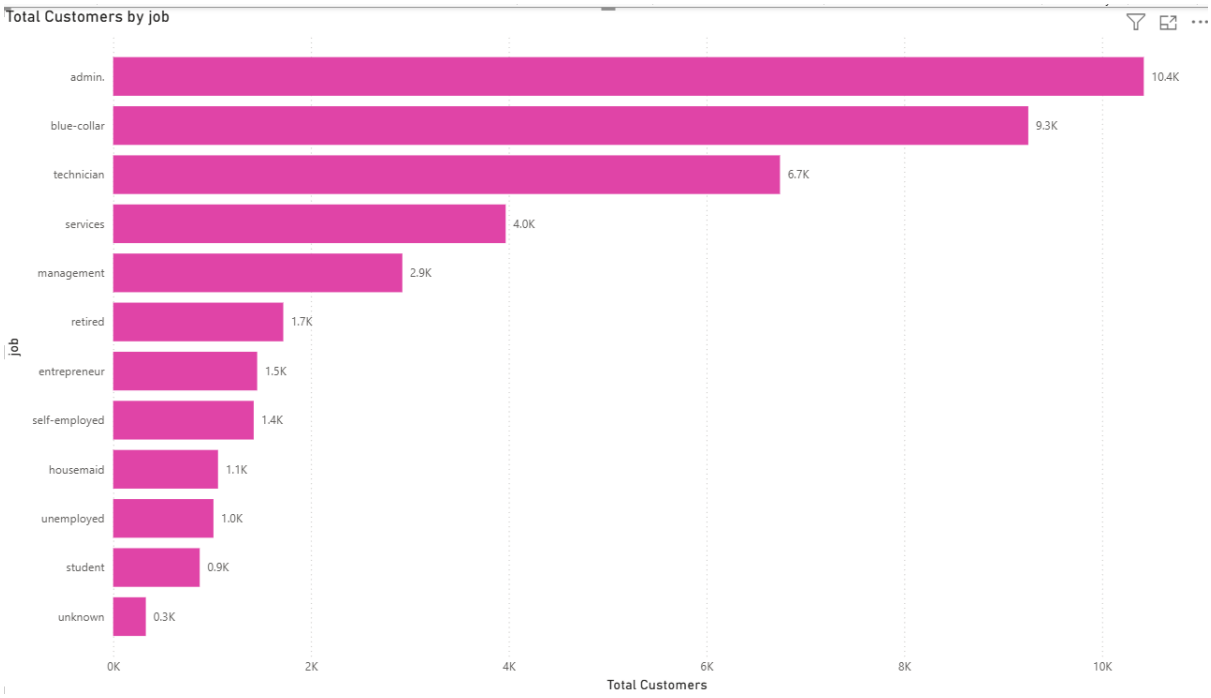
Creating DAX Measures for Analytics

Two new measures *Total Customers* and *Subscription Rate* are created.

Total Customers = COUNTROWS(WorkSheet)

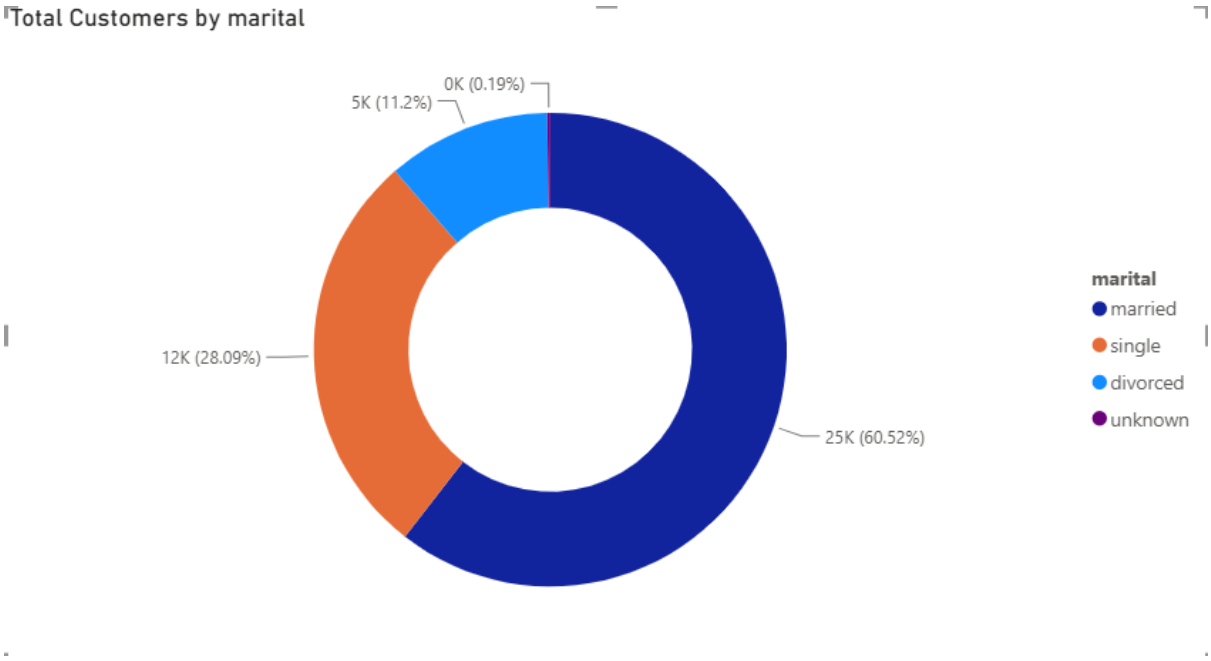
Subscription Rate = DIVIDE(CALCULATE(COUNTROWS(Worksheet), Worksheet[y] = "yes"), [Total Customers], 0)

(a) What is the distribution of the job type of the customers?



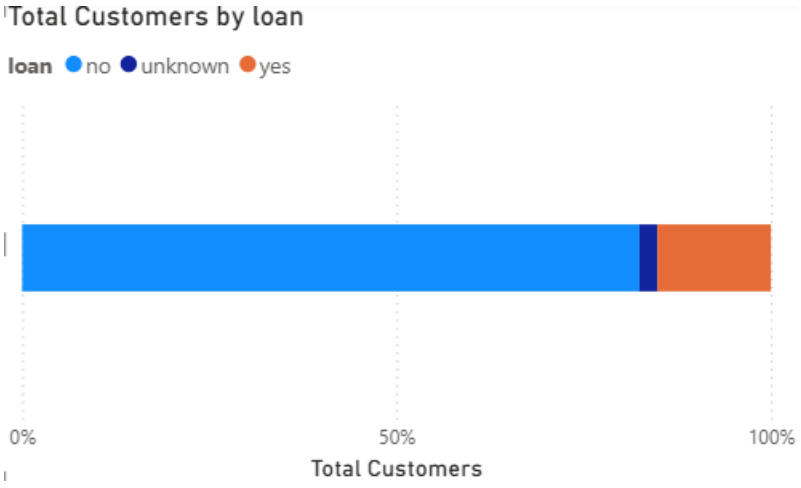
The chart shows that the bank's customer base is heavily dominated by administrative professionals, blue-collar workers, and technicians, who represent the largest occupational segments. Administrative staff make up the single largest group at 10.4K customers, followed closely by blue-collar workers at 9.3K and technicians at 6.7K. Mid-tier occupational categories include services and management, while sectors like retirees, entrepreneurs, and the self-employed comprise a smaller portion of the audience. Students and individuals with unknown job types form the smallest target groups, indicating that the bank's marketing campaigns primarily focus on steady-income employment segments.

(b) What is the distribution of customers by marital status?



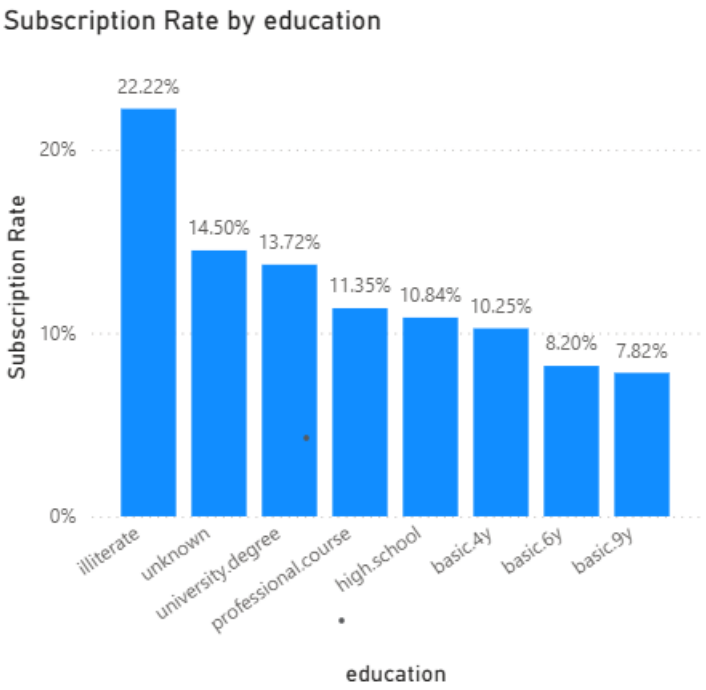
The donut chart reveals that married individuals form the clear majority of the bank's customer base, accounting for 60.52% of the total, which translates to roughly 25K individuals. Single customers represent the second largest demographic segment, comprising 28.09% or about 12K of the targeted population. Divorced individuals account for a much smaller share at 11.2% with a count of 5K, while those with an unknown status make up a negligible fraction at just 0.19%. This distribution indicates that the bank's marketing outreach is overwhelmingly centered around married and single individuals, who collectively make up nearly 89% of all records.

(c) What percentage of customers have a personal loan?



The 100% stacked bar chart shows that the vast majority of targeted bank customers do not currently hold a personal loan. Looking at the horizontal distribution, the segment representing individuals without a personal loan spans across roughly 82% of the entire customer base. In contrast, customers who actively hold a personal loan make up a significantly smaller portion, accounting for approximately 16% of the chart. A tiny, negligible fraction of the audience falls under an unknown status, indicating that the bank's marketing campaign is heavily directed toward individuals who are relatively free of short-term personal debt liabilities.

(d) What is the subscription rate based on education level?



The illiterate category has the highest subscription rate at 22.22%, indicating they are the most likely to subscribe. Customers with unknown education and university degrees also show relatively high subscription rates of 14.50% and 13.72%, respectively. Subscription rates generally decrease among customers with basic education levels, with basic.9y having the lowest rate at 7.82%. This suggests that education level influences subscription behavior, but other factors may also contribute to customers' decisions.

(e) Which marital group has the highest subscription rate?

marital	Subscription Rate
divorced	8.97%
married	9.21%
single	14.22%
unknown	7.14%
Total	**10.84%**

The table shows that the single marital status group achieves the highest overall conversion, leading with a subscription rate of 14.22%. Married and divorced customers demonstrate lower conversion metrics, yielding close subscription rates of 9.21% and 8.97% respectively. The unknown category exhibits the lowest performance, registering a subscription rate of just 7.14%. This variance indicates that single individuals show a significantly higher propensity to subscribe to term deposits compared to any other marital segment, dragging the collective average baseline to 10.84%.

Full Dashboard

